

Fran Menéndez Moya — Principal Software Engineer & Technical Architect

📞 +34 678 77 91 50 • ✉ fmenendezmoya@gmail.com • 🌐 fmenemo
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Executive Summary

Principal Software Engineer with 8+ years architecting high-scale platforms serving 2M+ users across consumer technology, mobility, and e-commerce sectors. Expert in performance optimization, enterprise migrations, and innovative infrastructure solutions. Proven track record scaling engineering teams, reducing operational costs, and delivering technical solutions that drive significant business growth and competitive advantages.

Core Technical Leadership

Platform Architecture: Enterprise-scale system design • Microservices architecture • Performance engineering • Zero-downtime migrations

Team Leadership: Cross-functional team management • Technical mentoring • Engineering standards • Hiring & retention

Performance Excellence: Query optimization • Caching strategies • CDN architecture • Infrastructure scaling

Innovation: Industry-first solutions • A/B testing infrastructure • Cross-platform integration • Technical strategy

Technical Expertise

Frontend (Expert): Next.js • React • Vue.js • TypeScript • Progressive Web Apps

Backend (Expert): Node.js • NestJS • Microservices • RESTful APIs • Server Functions

Cloud & Infrastructure: AWS (Cognito, Lambda) • Kubernetes • Docker • CDN Optimization • Message Queues

Data & Performance: PostgreSQL • MongoDB • Query Optimization • Caching • Real-time Systems

DevOps & Deployment: CI/CD • Container Orchestration • Zero-Downtime Deployments • Infrastructure as Code

Professional Experience

The Knot Worldwide

Principal Software Engineer

Remote

Apr 2025–Present

Leading technical architecture for consumer platform serving 2M+ weekly users

- **Enterprise Migration Leadership:** Architected PayloadCMS v2→v3 migration for 100,000+ products with zero downtime, reducing query response times from 850ms to 34ms (96% improvement) through strategic database optimization and caching architecture
- **Cross-Platform Authentication:** Designed company-wide session management system using AWS Cognito and microservices, enabling seamless user experience across 5+ products and reducing authentication latency by 78%
- **Infrastructure Optimization:** Implemented advanced caching strategies and query optimization serving 2M+ weekly users, reducing infrastructure costs by 40% while improving platform reliability to 99.98% uptime
- **Technical Strategy:** Established architectural standards adopted across 3 engineering teams, positioned platform for next-generation product launches with projected 300% user growth capacity

The Knot Worldwide

Lead Software Engineer

Remote

Oct 2023–Mar 2025

Drove platform evolution and engineering team growth for consumer technology leader

- **Revolutionary Infrastructure:** Created industry-first CDN-compatible A/B testing system, enabling experimentation for millions of users where technical limitations previously made it impossible, increasing conversion rates by 23%
- **Deployment Excellence:** Transformed release pipeline from 2+ hours to 1 minute (99.2% improvement), enabling daily deployments and reducing developer cycle time by 85%
- **E-commerce Platform:** Led development of integrated commerce solution, expanding platform capabilities and creating new revenue streams projected at \$2M+ annually
- **Team Leadership:** Managed 8-person cross-functional engineering team, established technical standards, mentored 5 developers resulting in 2 promotions and 100% team retention

MOBIKO GmbH

Team Lead & Architecture

Berlin, Germany

Mar 2023–Sep 2023

Led technical team and business growth for European mobility platform

- **Business Growth:** Drove 20% customer acquisition growth through strategic platform improvements and technical decision-making, contributing to \$1.2M ARR increase
- **Enterprise Integration:** Enabled partnerships with 3 global enterprises through scalable microservices architecture, processing 500k+ daily mobility events
- **Team Scaling:** Expanded engineering team by 20% while maintaining code quality standards and improving delivery velocity by 35%
- **Technical Architecture:** Designed event-driven microservices using NestJS and Kubernetes, handling real-time mobility data with 99.9% processing accuracy

MOBIKO GmbH

Senior Full-Stack Developer

Berlin, Germany

Aug 2020–Mar 2023

Scaled platform from startup to enterprise serving 10,000+ active users

- **Massive Scale Growth:** Scaled user base from 3,000 to 10,000+ users (233% growth) through performance optimization and strategic feature development
- **Performance Engineering:** Reduced API response times by 72% and page load times by 70% through caching optimization and database query improvements
- **DevOps Transformation:** Automated deployment pipeline reducing release time from 3 days to instantaneous, enabling continuous deployment and improving developer productivity by 60%
- **Technical Foundation:** Built scalable architecture using Vue.js, NestJS, and PostgreSQL in Kubernetes environment, supporting 50k+ daily API calls

Hiberus Tecnología

Senior Full-Stack Developer

Zaragoza, Spain

Jan 2020–Jul 2020

Optimized high-traffic ticketing platform for sports and entertainment

- **Revenue Impact:** Increased ticketing sales by 40% through real-time optimization and UX improvements, contributing to \$800k additional revenue
- **Cost Optimization:** Reduced platform maintenance costs by 95% through architectural redesign and automation, saving \$120k annually
- **User Engagement:** Improved user engagement metrics by 4x through real-time features and API performance optimization
- **External Integration:** Successfully integrated La Liga's official API, enabling real-time sports data for 100k+ concurrent users

Hiberus Tecnología

Full-Stack Developer

Zaragoza, Spain

Nov 2017–Jan 2020

Built high-capacity platform handling 1M+ daily transactions

- **Enterprise Scale:** Engineered platform processing 1M+ daily entries with real-time synchronization across 5 geographic locations, maintaining 99.95% uptime
- **Cost Efficiency:** Reduced API operational costs by 70% through intelligent caching and query optimization, saving \$200k annually
- **Multi-Platform Architecture:** Developed cross-platform solution using Angular, Ionic, and NestJS with Firebase integration supporting web, iOS, and Android

Key Achievements

Performance Excellence: Consistently delivered 70-96% performance improvements across multiple platforms and technologies

Scale Leadership: Grew user bases by 233% while maintaining platform stability and team performance

Cost Optimization: Achieved 95% maintenance cost reductions through strategic architectural decisions

Innovation Leadership: Created industry-first CDN-compatible A/B testing infrastructure serving millions of users

Team Development: Successfully scaled engineering teams by 20%+ with 100% retention rates

Recognition & Awards

Technical Innovation: NASA Space Apps Global Finalist & Galactic Impact Award • Google Hash Code Winner
ImagineCode Blockchain Challenge Winner • Multiple Innovation Technology Awards

Industry Impact: Recognized for breakthrough solutions in performance optimization and large-scale platform architecture

Languages & Education

Languages: Spanish (Native) • English (Advanced - Cambridge Certified C1)

Education: Computer Engineering, Universidad de Zaragoza • Software Engineering Specialization